Subject: / /

Transmission medium

↳ transmission b/w two devices

Physical layer — Physical layer

TM → Cables / Wired → Guided

wireless ↳ Unguided.

Twist → to reduce noise

T.M. ↳ Channel / Prop.

↳ 1. Speed

→ LAN

→ TV cables

→ Optical fibre

Subject: / /

* Transmission Medium :-

↳ In which data is transferred from one node to another

↳ It provide a pathway over which message can travel from sender to reciever

↳ Each of the message can be send in the form of data by converting them into the bits / binary

↳ These pathways can be wired, or wireless

↳ The choice of medium depend on factor like → Distance & Speed.

Transmission Media

- Guided Media (wired)
  - Co-axial
    1. Baseband
    2. Broad band
  - Twisted
    1. UTP
    2. STP
  - Fiber Optics
- (wireless) Unguided Media
  - Infra red
  - Micro waves.
  - Radio waves

Subject: Guided Media

↳ It consist of wire - data transfer

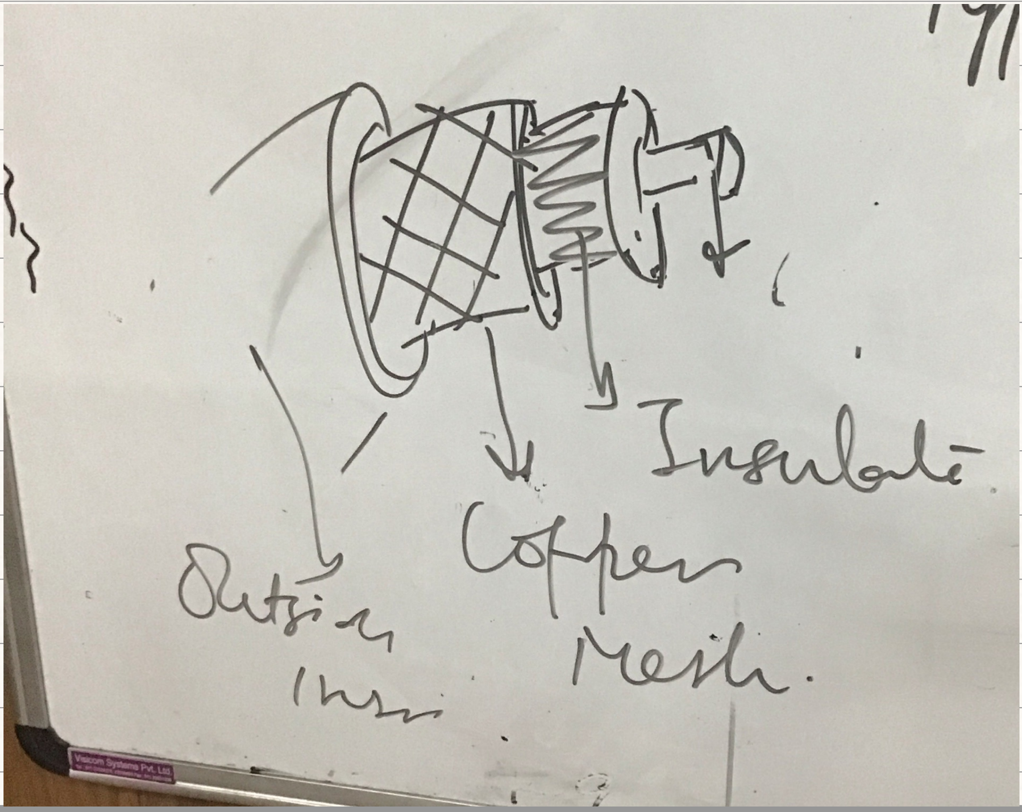
→ It is a physical link b/w client and server
→ It is used for shorter distance
→ It provide high speed of data.

↳ 1. Co-axial → Baseband
→ Broadband

↳ Inner conductor made up of copper & outer conductor made up of mesh.

↳ Middle core made up of non-conductive cover

→ It has a higher freq. as compared to twisted paired cable.



Subject: 2. Twisted

Is Pair of copper cable twisted with each other

↳ It has two seperately insulated conductor wires.

1. UTP (Unshielded Twisted pair)

→ Mostly used in telecomm.

→ low cost, simple to install & high speed.

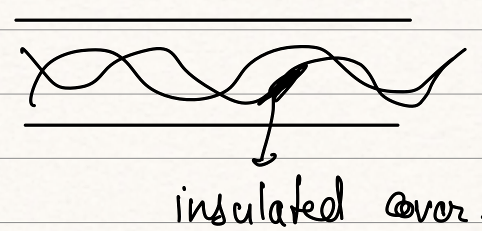
→ It is categorize b/w 4 - 200 mbps.

2. STP (Shielded T. P.)

→ Mostly used in ethernet, medical equipments

→ high data transmission rate.

→ installation easy & expensive.



Subject: 3. Fiber Optics :-

→ It used to send the data by pulses of light

→ It perform well in long distance data

→ Carry large amount of information, and transmit data at a high speed.

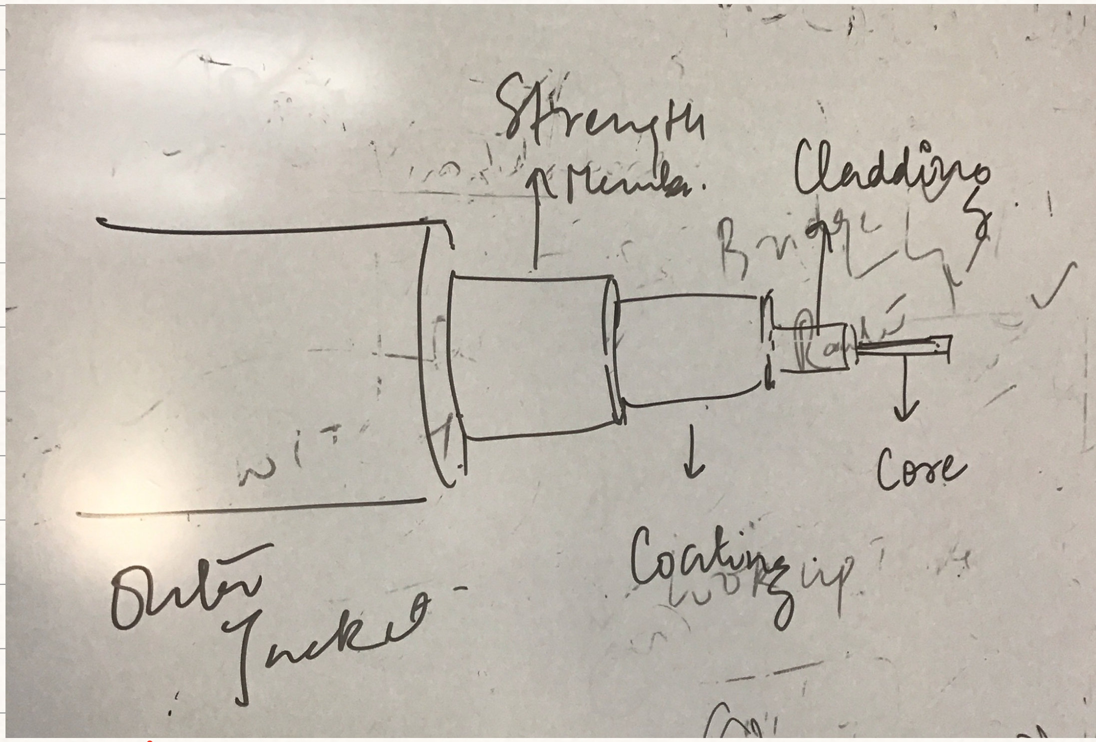
→ High cost, difficult for maintance & installation.

1. Core → Design to transport light
2. Cladding → A thin layer that transfer data through the fiber
3. Coating → Further protection the fiber core

4. Strength Member :- keep fibre core from damage or installation.

5. Outer Jacket
↳ is designed to protect cable from enviromental damage.

Subject: → internet, computer network
military.



2. Unguided Medium :- Unguided Transmission Media also reffred as wireless media,

→ No physical medium is required
→ It is used for larger distance
→ Broadcast through air
→ Less secure.

I. Radio waves :-
Type of E.M. wave, which are transmitted in every direction of free Space
→ Is used in wide area network
→ This wave has a 3kHz to 1GHz frequency

range
Subject: → Higher data transmission rate.

→ It is useful for multicasting
↳ 1 client & multiple server.

→ Application
↳ FM radio, Mobile

2. Macrowave :- Uses antenna as a main element for sending and recieving data

→ Distance covered by the signal is proportional to the height of antenna.

→ Range 1 GHz to 300 GHz

→ Carry large amount of information with capacity

Appli. → Statellite network.

3. Infra red :- It is a wireless technology used for Communication over extremely short distence.

→ Freq. range is from 300 GHz to 400 Tera Hz.

→ Data cannot passe through the obstacles, → it provide high band width & security.

→ it holds lower information rate of transmission.

Subject:

Appli:- TV remote, wireless mouse
keyboard